

Supporting Information

Transition metal-doped two-dimensional TCNE as efficient bifunctional ORR/OER electrocatalysts: A first principles systematic screening study

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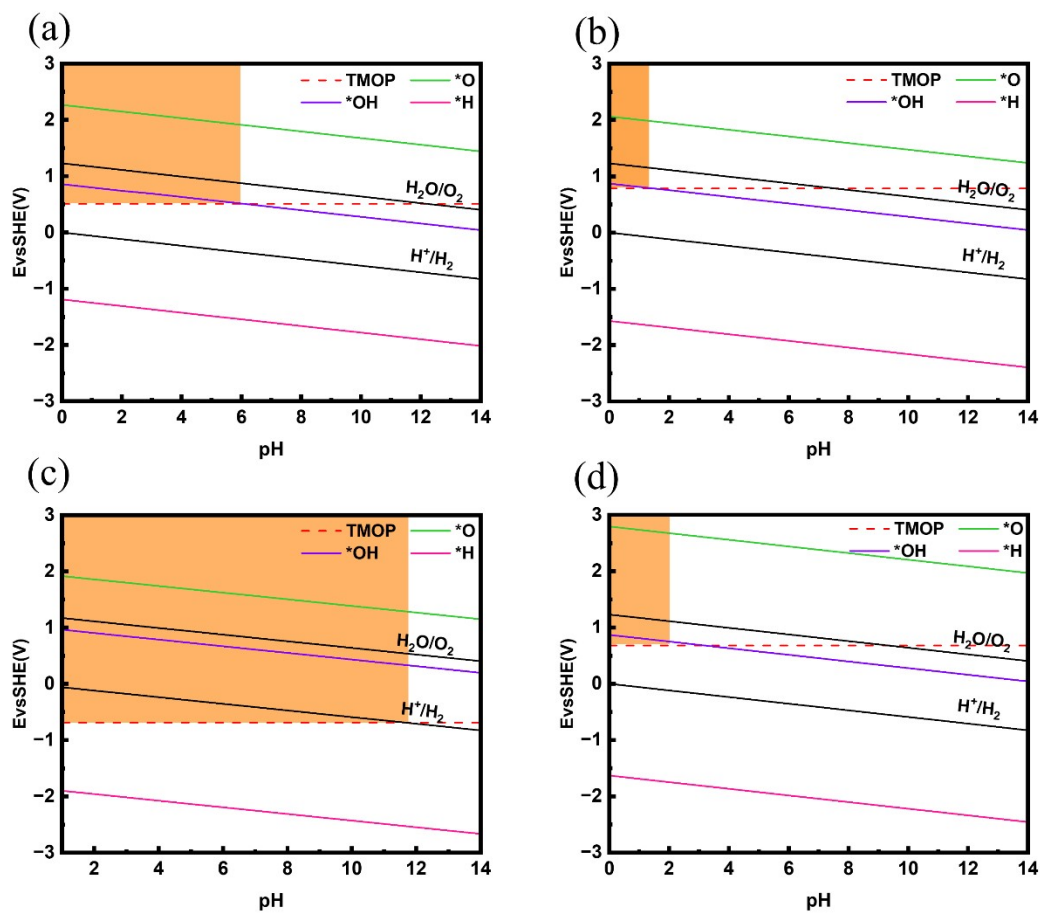


Figure S1 Surface Pourbaix diagrams for the electrochemical stability of (a) Ru-pTCNE, (b) Os-pTCNE, (c) Mn-cTCNE and (d) Os-cTCNE.

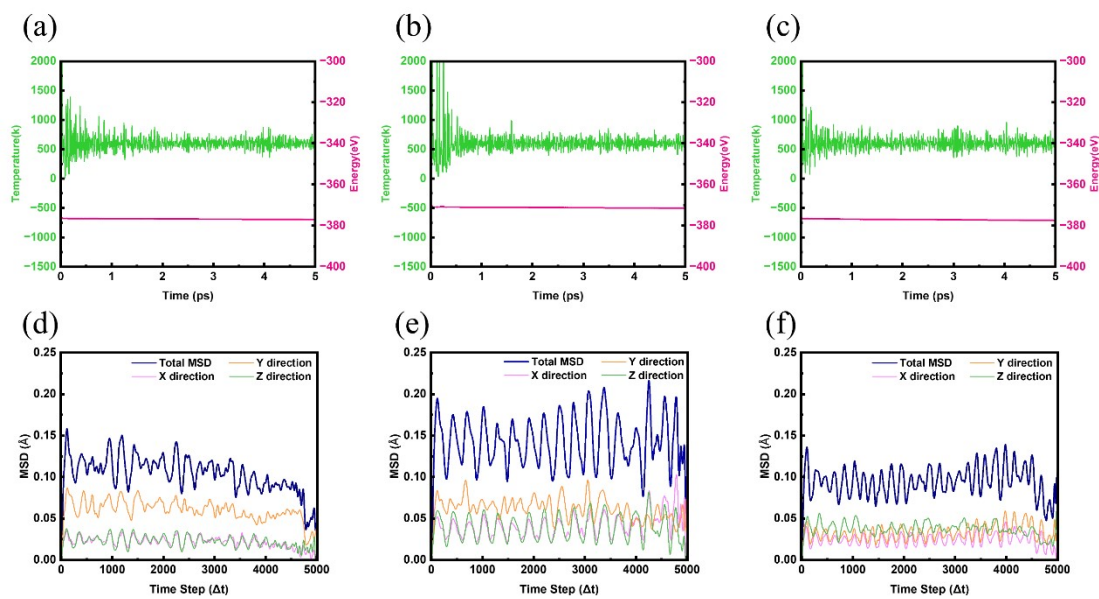


Figure S2 the AIMD and mean-square deviation (MSD) for the (a, d) Os-pTCNE, (b, e) Ru-pTCNE, and (c, f) Os-cTCNE.

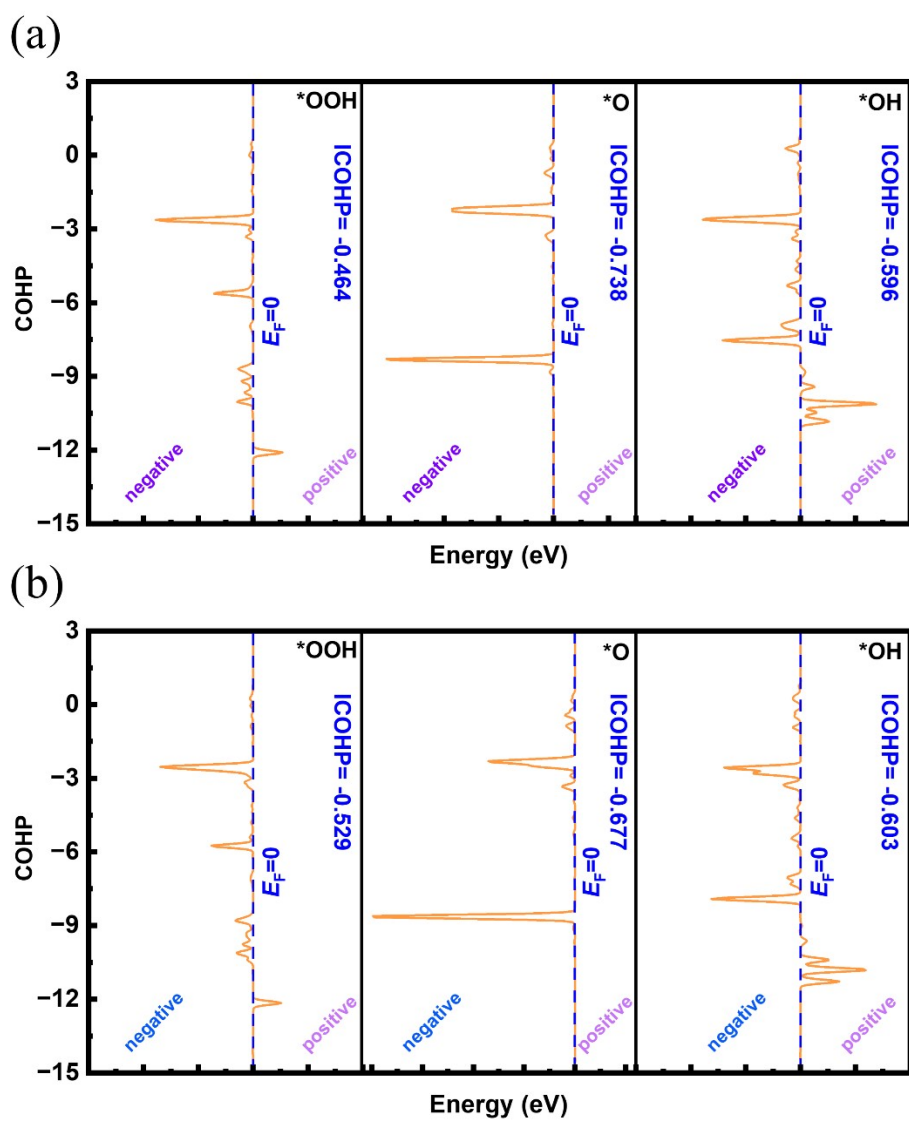


Figure S3 the COHP between the TM and O atoms : (a) Ru-pTCNE, and (b) Os-cTCNE.

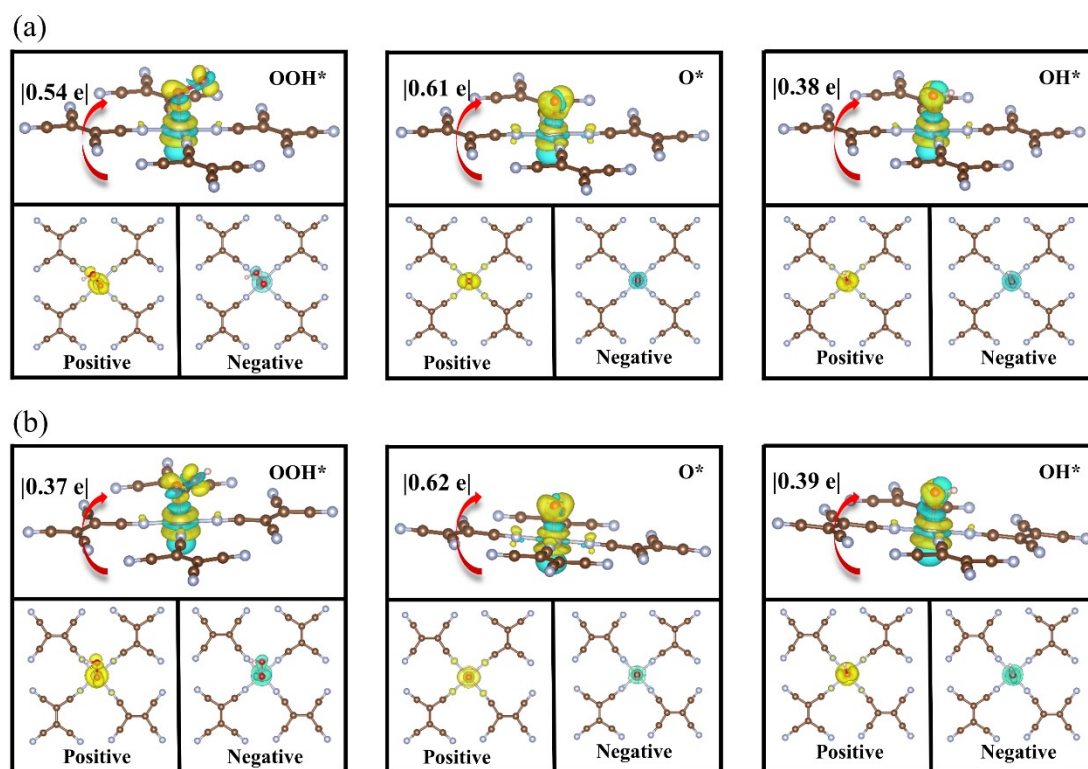


Figure S4 the differential charge density diagram of intermediates adsorbed on (a) Ru-pTCNE, and (b) Os-cTCNE substrate.

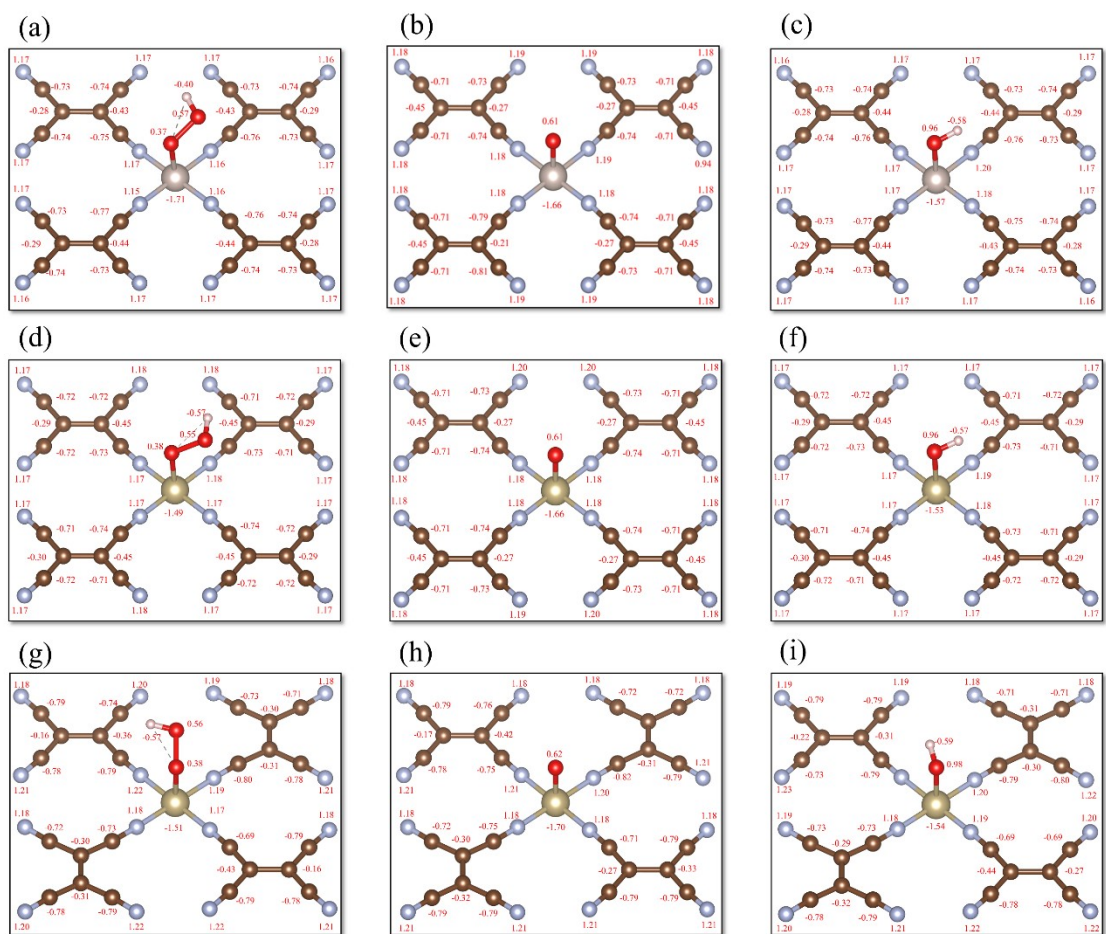


Figure S5 the Bader charge diagram of intermediates adsorbed on (a-c) Ru-pTCNE, (d-f) Os-pTCNE, and (g-i) Os-cTCNE substrate.

Table S1 Adsorption energies of O₂ on TM-cTCNE

TM-cTCNE	E_{tot}	$E_{\text{TM-TCNE-c}}$	E_{O_2}	E_{ads}
Sc-cTCNE	-382.530	-371.145	-9.860	-1.525
Ti-cTCNE	-385.316	-373.618	-9.860	-1.838
V-cTCNE	-387.052	-375.907	-9.860	-1.285
Cr-cTCNE	-387.592	-376.992	-9.860	-0.740
Mn-cTCNE	-384.568	-374.2226	-9.860	-0.442
Fe-cTCNE	-379.452	-368.973	-9.860	-0.619
Co-cTCNE	-373.893	-363.955	-9.860	-0.079
Ni-cTCNE	-365.624	-355.882	-9.860	0.118
Zn-cTCNE	-350.586	-340.569	-9.860	-0.157
Y-cTCNE	-383.272	-371.729	-9.860	-1.680
Zr-cTCNE	-387.374	-377.461	-9.860	-0.052
Nb-cTCNE	-389.316	-377.085	-9.860	-2.372
Mo-cTCNE	-388.546	-377.114	-9.860	-1.562
Ru-cTCNE	-380.807	-370.146	-9.860	-0.801
Rh-cTCNE	-373.627	-364.372	-9.860	0.605
Pd-cTCNE	-362.764	-352.887	-9.860	-0.017
Ag-cTCNE	-347.665	-337.786	-9.860	-0.019
Cd-cTCNE	-347.106	-337.102	-9.860	-0.144
Ta-cTCNE	-394.798	-382.532	-9.860	-2.406
W-cTCNE	-394.327	-382.682	-9.860	-1.785
Re-TCNE	-392.624	-379.555	-9.860	-3.209
Os-cTCNE	-386.502	-376.053	-9.860	-0.589
Ir-cTCNE	-378.795	-369.455	-9.860	0.520
Pt-cTCNE	-367.624	-357.735	-9.860	-0.029
Au-cTCNE	-351.525	-342.760	-9.860	1.096

Table S2 Adsorption energies of O₂ on TM-pTCNE

TM-TCNE-p	E_{tot}	$E_{\text{TM-TCNE-c}}$	E_{O_2}	E_{ads}
Sc-TCNE-p	-384.514	-372.550	-9.860	-2.104
Ti-TCNE-p	-386.641	-374.757	-9.860	-2.024
V-TCNE-p	-388.360	-377.118	-9.860	-1.383
Cr-TCNE-p	-388.953	-378.268	-9.860	-0.825
Mn-TCNE-p	-385.885	-375.484	-9.860	-0.541
Fe-TCNE-p	-380.775	-370.360	-9.860	-0.555
Co-TCNE-p	-375.228	-365.196	-9.860	-0.171
Ni-TCNE-p	-367.198	-357.282	-9.860	-0.057
Cu-TCNE-p	-358.091	-348.062	-9.860	-0.169
Zn-TCNE-p	-351.841	-341.818	-9.860	-0.163
Y-TCNE-p	-384.700	-373.136	-9.860	-1.704
Zr-TCNE-p	-388.836	-376.548	-9.860	-2.428
Nb-TCNE-p	-390.683	-378.375	-9.860	-2.448
Mo-TCNE-p	-389.968	-378.425	-9.860	-1.682
Ru-TCNE-p	-382.255	-371.493	-9.860	-0.902
Rh-TCNE-p	-375.960	-365.682	-9.860	-0.418
Pd-TCNE-p	-364.115	-354.176	-9.860	-0.079
Ag-TCNE-p	-352.520	-342.577	-9.860	-0.084
Cd-TCNE-p	-348.413	-338.362	-9.860	-0.191
Hf-TCNE-p	-394.138	-381.939	-9.860	-2.339
Ta-TCNE-p	-396.370	-384.015	-9.860	-2.494
W-TCNE-p	-395.793	-384.016	-9.860	-1.917
Re-TCNE-p	-392.485	-380.978	-9.860	-1.647
Os-TCNE-p	-387.922	-377.500	-9.860	-0.562
Ir-TCNE-p	-380.815	-370.865	-9.860	-0.090
Pt-TCNE-p	-369.202	-359.253	-9.860	-0.089
Au-TCNE-p	-353.810	-343.876	-9.860	-0.074

Table S3 the COHP and d_{TM-O} of intermediates adsorbed on TM-TCNE substrates.

	OOH*		O*		OH*	
	COHP	d _{TM-O} (Å)	COHP	d _{TM-O} (Å)	COHP	d _{TM-O} (Å)
Ru-pTCNE	-0.464	1.990	-0.738	1.830	-0.596	1.987
Os-pTCNE	-0.521	1.984	-0.665	1.804	-0.641	1.970
Os-cTCNE	-0.529	1.989	-0.677	1.831	-0.603	1.988

POSCAR of Os-pTCNE

Os-pTCNE

1.0000000000000000

14.2174484714603757 0.0000000000000000 0.0050555567185611

0.0000000000000000 18.6112873865244701 0.0000000000000000

-0.0077466261103633 0.0000000000000000 13.9575839286109584

Os N C

4 16 24

Direct

-0.0000000000000000 0.5000000000000000 -0.0000000000000000

0.5000000000000000 0.5000000000000000 -0.0000000000000000

-0.0000000000000000 0.5000000000000000 0.5000000000000000

0.5000000000000000 0.5000000000000000 0.5000000000000000

0.4059770838849686 0.5000000000000000 0.1001458617207674

0.4058780446351707 0.5000000000000000 0.4000094343246532

0.0940229161150314 0.5000000000000000 0.3998541382792326

0.0941219553648293 0.5000000000000000 0.0999905656753468

0.9059770838849686 0.5000000000000000 0.1001458617207674

0.9058780446351706 0.5000000000000000 0.4000094343246532

0.5940229161150314 0.5000000000000000 0.3998541382792326

0.5941219553648294 0.5000000000000000 0.0999905656753468

0.4059770838849686 0.5000000000000000 0.6001458617207674

0.4058780446351707 0.5000000000000000 0.9000094343246532

0.0940229161150314 0.5000000000000000 0.8998541382792326

0.0941219553648293 0.5000000000000000 0.5999905656753468

0.9059770838849686 0.5000000000000000 0.6001458617207674

0.9058780446351706 0.5000000000000000 0.9000094343246532

0.5940229161150314 0.5000000000000000 0.8998541382792326

0.5941219553648294 0.5000000000000000 0.5999905656753468

0.3536714214381031 0.5000000000000000 0.1661324361820112

0.3535383738770202 0.5000000000000000 0.3340515422405664

0.3011067902527997 0.5000000000000000 0.2500375400769185

0.1463285785618969 0.5000000000000000 0.3338675638179888

0.1464616261229798 0.5000000000000000 0.1659484577594335

0.1988932097472003 0.5000000000000000 0.2499624599230815

0.8536714214380960 0.5000000000000000 0.1661324361820112

0.8535383738770203 0.5000000000000000 0.3340515422405664

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0.6463285785619040 0.5000000000000000 0.3338675638179888

0.6464616261229797 0.5000000000000000 0.1659484577594335

0.6988932097472003 0.5000000000000000 0.2499624599230815

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0.3535383738770202 0.5000000000000000 0.8340515422405665

0.3011067902527997 0.5000000000000000 0.7500375400769185

0.1463285785618969	0.5000000000000000	0.8338675638179889
0.1464616261229798	0.5000000000000000	0.6659484577594335
0.1988932097472003	0.5000000000000000	0.7499624599230815
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0.8011067902527997	0.5000000000000000	0.7500375400769185
0.6463285785619040	0.5000000000000000	0.8338675638179889
0.6464616261229797	0.5000000000000000	0.6659484577594335
0.6988932097472003	0.5000000000000000	0.7499624599230815

POSCAR of Ru-pTCNE

Ru-pTCNE

1.0000000000000000

14.1614225971045666 0.0000000000000000 0.0192358777408655

0.0000000000000000 18.3986358598505184 0.0000000000000000

-0.0068844546371356 0.0000000000000000 14.0397584346517448

Ru N C

4 16 24

Direct

-0.0000000000000000 0.5000000000000000 -0.0000000000000000

0.5000000000000000 0.5000000000000000 -0.0000000000000000

-0.0000000000000000 0.5000000000000000 0.4999999899999999

0.4999999930000030 0.5000000000000000 0.4999999899999999

0.4058801863019376 0.5000000000000000 0.1006910850770527

0.4048373669840679 0.5000000000000000 0.4004360708150402

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0.3535079173399890 0.5000000000000000 0.3342484980391938

0.3009608689259818 0.5000000000000000 0.2502950854871013

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0.8535079493399846	0.5000000000000000	0.3342484980391938
0.8009608369259862	0.5000000000000000	0.2502950854871013
0.6455717206973735	0.5000000000000000	0.3331420093756024
0.6464920116600157	0.5000000000000000	0.1657514739608038
0.6990391880740194	0.5000000000000000	0.2497048865128964
0.3544282333026227	0.5000000000000000	0.6668579346243929
0.3535079433399840	0.5000000000000000	0.8342485590391955
0.3009608629259813	0.5000000000000000	0.7502951294871050
0.1455717626973769	0.5000000000000000	0.8331420343756010
0.1464920366600143	0.5000000000000000	0.6657514099608055
0.1990391330740184	0.5000000000000000	0.7497048405128997
0.8544282333026227	0.5000000000000000	0.6668579346243929
0.8535079433399841	0.5000000000000000	0.8342485590391955
0.8009608949259839	0.5000000000000000	0.7502951294871050
0.6455717146973730	0.5000000000000000	0.8331420343756010
0.6464920046600116	0.5000000000000000	0.6657514099608055
0.6990391810740153	0.5000000000000000	0.7497048405128997

POSCAR of Os-cTCNE

Os-cTCNE

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14.0829360728130215	-0.0026170549613789	0.0000000000000000
-0.0026172610041420	14.0916668008472801	0.0000000000000000
0.0000000000000000	0.0000000000000000	17.1828986755860349

Os	N	C
4	16	24

Direct

0.0003511008743828	0.9944640194448340	0.5000000000000000
0.9998781246377462	0.4942095682171441	0.5000000000000000
0.4998703764513692	0.9942255168369627	0.5000000000000000
0.5003503412153165	0.4944723973654277	0.5000000000000000
0.1025634369211375	0.0861117819143828	0.5000000000000000
0.1021785368457415	0.4028206968382548	0.5000000000000000
0.0915028881489803	0.5963652894755924	0.5000000000000000
0.0919372156964045	0.8921373218379358	0.5000000000000000
0.4083773780049498	0.8919615139840431	0.5000000000000000
0.6022317972924682	0.9028802462060810	0.5000000000000000
0.8981629706100165	0.9029970291834672	0.5000000000000000
0.8977024219935146	0.5858917099178669	0.5000000000000000
0.9083298380314773	0.3919598540302331	0.5000000000000000

0.9087179133323247	0.0965032677155548	0.5000000000000000
0.5914368111515955	0.0964176740335816	0.5000000000000000
0.3976509593537187	0.0858507551397949	0.5000000000000000
0.3981217887849127	0.4030163058171112	0.5000000000000000
0.5919048428078213	0.3921555628447635	0.5000000000000000
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0.3330936749895557	0.1396273053386494	0.5000000000000000
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0.7502447252134559	0.6931238001573234	0.5000000000000000
0.8331466423739826	0.6396733162357862	0.5000000000000000
0.6672818445119333	0.6397592549418221	0.5000000000000000
0.6451150315372871	0.3271527209716822	0.5000000000000000
0.8550004304536927	0.3270235928364897	0.5000000000000000
0.8016991696505400	0.2440557669398018	0.5000000000000000
0.6983419954597991	0.2441357485654834	0.5000000000000000
0.6449065340353286	0.1612446303911597	0.5000000000000000
0.8550694933225316	0.1611467305040932	0.5000000000000000